Short paper proposal

On the Estimation Window and Horizon Dynamics of the HAR Model in Cryptocurrency Volatility

Diego Astaburuaga[a,b]

[a]*Departamento de Industrias, Universidad Técnica Federico Santa María*
[b]*Departamento de Matemáticas, Universidad Técnica Federico Santa María*

Profesor: Werner Kristjanpoller.

Abstract—
The Heterogeneous Autoregressive (HAR) model is widely used for modeling and forecasting realized volatility due to its ability to capture volatility persistence across multiple time horizons. Despite its extensive use in empirical finance, practical implementation choices such as the selection of the rolling estimation window are often treated as ad hoc decisions rather than as empirical questions.
This paper investigates the impact of the estimation window size on the performance and interpretation of the HAR model in cryptocurrency markets. Using daily realized volatility from major cryptocurrencies, we evaluate a range of rolling window configurations and analyze how the resulting coefficients evolve across different horizons. The results show that the forecasting performance of the HAR model is sensitive to the estimation window length, with optimal windows typically ranging between approximately 700 and 1200 observations, corresponding to two to three years of daily data.
In addition, the analysis reveals that the relative importance of the daily, weekly, and monthly components varies over time and depends on the estimation horizon. While the daily and weekly components often alternate in importance across regimes, the monthly component becomes increasingly relevant as the estimation window grows, reflecting the long-memory nature of volatility. These findings suggest that the empirical interpretation of HAR coefficients is inherently linked to the estimation design.
Overall, the results highlight the importance of carefully selecting the estimation window when implementing HAR models and provide new evidence on the dynamic structure of volatility horizons in cryptocurrency markets.

Keywords—*Realized volatility, HAR model, Cryptocurrency markets, Volatility persistence, Rolling window estimation*

1. Introduction

Forecasting financial market volatility is a central problem in asset pricing, risk management, derivative valuation, and portfolio allocation. Accurate volatility forecasts are essential for measuring risk exposure, computing Value-at-Risk (VaR), pricing derivatives, and designing volatility-timing strategies. With the increasing availability of high-frequency data, volatility modeling has progressively shifted from latent conditional variance models toward observable measures constructed from intraday returns. Among these, Realized Volatility (RV) has emerged as a standard proxy for daily volatility.

The theoretical foundation of RV is rooted in continuous-time finance. Under suitable regularity conditions, realized measures consistently estimate quadratic variation, providing an asymptotically error-free proxy for integrated variance as sampling frequency increases [2]. This development represented a structural advance in volatility modeling, as it allowed researchers to work with observable volatility proxies rather than relying exclusively on latent processes such as GARCH-type models.

Building on this framework, [5] introduced the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), inspired by the Heterogeneous Market Hypothesis [1]. The central idea is that market participants operate at different time horizons—short-term traders, medium-term investors, and long-term institutions—creating a cascade mechanism in volatility dynamics. The HAR-RV model formalizes this intuition through a simple additive specification:

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \varepsilon_{t+1}, \tag{1}$$

where RV_t denotes daily realized volatility, while $RV_{w,t}$ and $RV_{m,t}$ represent weekly and monthly averages, respectively. Despite its linear form, the HAR model successfully reproduces long-memory behavior and strong volatility persistence, and has become a benchmark model across asset classes.

While HAR has been extensively studied in equity, commodity, and energy markets, its application to cryptocurrency markets raises structural questions. Cryptocurrency markets operate continuously (24/7), exhibit frequent regime shifts, experience pronounced speculative episodes, and undergo rapid structural transformations in liquidity and participation. These characteristics suggest that volatility dynamics may be more unstable and regime-dependent than in classical markets.

Two interconnected issues become particularly relevant in this context. First, the estimation window size introduces a trade-off between statistical precision and regime adaptability. Larger windows reduce estimation variance but may dilute local structural dynamics. Smaller windows adapt more quickly to regime changes but increase parameter uncertainty. This trade-off is amplified in cryptocurrency markets, where structural breaks and volatility clustering are frequent.

Second, although the HAR framework decomposes volatility into daily, weekly, and monthly components, relatively limited attention has been devoted to studying how the relative importance of these horizons evolves over time. In highly dynamic environments such as cryptocurrencies, it is plausible that short-term components dominate during turbulent regimes, while longer horizons regain importance during stable periods. Understanding this time-varying horizon importance is essential for interpreting the economic meaning of HAR coefficients in continuously traded markets.

This paper investigates the time-varying importance of HAR horizons in cryptocurrency markets. We examine how the predictive contribution of the daily, weekly, and monthly components evolves across time and volatility regimes. In addition, we study the sensitivity of the HAR model to rolling estimation window size, evaluating the balance between statistical efficiency and regime adaptability.

[This paper proceeds as follows. Section X describes the data and construction of realized volatility measures. Section X introduces the rolling-window estimation framework and window-size selection procedure. Section X defines the metrics used to quantify horizon

importance and presents the multi-asset evaluation strategy. Section X reports empirical findings, including window sensitivity analysis and time-varying coefficient dynamics. Section X concludes.]

2. Related Work

The use of high-frequency data to construct observable volatility measures represents a major development in financial econometrics. The seminal contributions of [2] formalized the theoretical underpinnings of realized volatility, showing that sums of intraday squared returns consistently estimate quadratic variation under mild assumptions.

However, high-frequency data are contaminated by microstructure noise. [3] demonstrate that sampling frequency involves a bias–variance trade-off, as ultra-high-frequency observations introduce measurement distortions. To address these issues, [4] propose realized kernel estimators that remain consistent in the presence of market frictions. Together, these contributions establish RV as a reliable and theoretically grounded proxy for daily volatility.

The HAR-RV model of [5] represents a parsimonious approximation to long-memory volatility dynamics. Motivated by the Heterogeneous Market Hypothesis [1], the model decomposes realized volatility into components associated with different time horizons.

Despite its linear structure, HAR captures key stylized facts of volatility, including persistence and multi-scale dependence. Its interpretability and empirical robustness have made it a standard benchmark in volatility forecasting.

Although conceptually simple, HAR performance depends critically on implementation details. [8] show that replacing OLS with weighted least squares (WLS) or robust regression can significantly improve forecast accuracy. Similarly, [9] provide theoretical justification for WLS estimation in multivariate HAR settings with heteroscedastic errors.

HAR models have been successfully applied across multiple markets, including agricultural commodities [12], crude oil [14], [20], [21], global equities [18], and renewable energy markets [15]. This broad applicability confirms HAR as a versatile and interpretable framework.

Cryptocurrency markets present unique volatility dynamics due to continuous trading, speculative intensity, and frequent structural shifts. Empirical evidence suggests that realized-volatility-based models outperform traditional daily-return-based approaches in this context.

[10] show that HAR models outperform GARCH-family models in forecasting Bitcoin volatility, particularly at short horizons. [16] introduce dynamic tuning strategies for HAR predictors in crypto markets, highlighting the importance of parameter adaptability. Furthermore, [22] compare HAR and machine learning approaches incorporating sentiment variables, finding that nonlinear models may better exploit exogenous information.

These findings suggest that while HAR remains a strong baseline, its static linear structure may be restrictive in rapidly evolving markets.

A large body of literature extends HAR to address nonlinearity, structural breaks, and time variation. Time-varying parameter models [7], [13] allow coefficients to evolve dynamically. Structural break and regime-switching HAR models [14], [19] incorporate nonlinear dynamics, often improving predictive performance.

Other extensions include jump-augmented HAR models [11], shrinkage-based high-dimensional HAR frameworks [18], Hidden Markov structures [12], and Markov time-varying transition models [15]. Hybrid econometric–machine learning models combine HAR with GRU, LSTM, SVM, or random forest architectures [20], [21], aiming to integrate interpretability with nonlinear predictive power.

Despite this rich literature, comparatively less attention has been devoted to directly studying the structural stability of HAR horizon contributions over time, particularly in continuously traded cryptocurrency markets. Most studies focus on enhancing predictive accuracy through additional variables or nonlinear structures, rather than analyzing how the relative importance of daily, weekly, and monthly components evolves across regimes.

The present study addresses this gap by systematically analyzing the time-varying importance of HAR horizons in cryptocurrency markets and evaluating the sensitivity of results to estimation window size. By focusing on horizon contribution rather than proposing an entirely new nonlinear structure, we aim to provide structural insight into whether the classical HAR decomposition retains its economic interpretation under 24/7 trading conditions.

3. Data

The data were obtained from the public Binance API. Five representative cryptocurrencies were selected for the empirical study. Table 1 summarizes their short names, complete names, and a brief description. The selection aims to include assets with high liquidity, long trading history, and heterogeneous structural characteristics within the cryptocurrency ecosystem.

The raw dataset consists of intraday observations sampled at a 5-minute frequency. The initial download spans from 01-01-2017 to 01-01-2026, subject to data availability for each asset. This horizon covers multiple distinct market regimes, including expansionary phases, severe drawdowns, and prolonged high-volatility episodes. A preliminary data quality assessment indicates no major inconsistencies. For each cryptocurrency, more than 95% of the downloaded intraday data present no issues. The remaining fraction corresponds to a limited number of isolated missing intraday observations, amounting to fewer than approximately 20 calendar days in total across assets. The data acquisition pipeline and validation procedures are documented in a separate technical report (available upon request).

The choice of a 5-minute sampling frequency follows the established realized volatility literature, which emphasizes the trade-off between information content and microstructure noise. Extremely high-frequency data (e.g., tick-level) may be contaminated by bid–ask bounce and market frictions, while excessively coarse sampling may discard relevant intraday variation. The 5-minute frequency has been widely adopted as a practical compromise in volatility estimation (see, e.g., [2], [3], [4]).

For each cryptocurrency, the `open time` field is used to ensure temporal alignment, and the corresponding `open price` is employed to compute log-returns. Daily realized volatility (RV) is constructed as the sum of squared intraday log-returns within each calendar day:

$$\mathrm{RV}_t = \sum_{i=1}^{M_t} r_{t,i}^2, \tag{2}$$

where $r_{t,i}$ denotes the i-th 5-minute log-return within day t, and M_t is the number of intraday intervals in that day.

Stylized Facts of Cryptocurrency Volatility

Cryptocurrency markets exhibit several well-documented stylized facts that motivate the use of flexible and high-frequency-based volatility models. First, returns display heavy tails and excess kurtosis, indicating a higher probability of extreme events relative to Gaussian benchmarks. Second, volatility clustering is prominent: large price movements tend to be followed by further large movements, consistent with persistence and long-memory behavior in volatility dynamics. Third, abrupt regime shifts and jump components are frequent, reflecting market-specific events, liquidity shocks, and structural breaks. These empirical properties justify the use of realized volatility measures derived from intraday data and support the exploration of nonlinear and hybrid modeling frameworks[1].

Asset Description

Table 1. Selected cryptocurrencies and basic description.

Short Name	Complete Name	Description
BTC	Bitcoin	Largest cryptocurrency by market capitalization; benchmark digital asset.
ETH	Ethereum	Smart-contract platform enabling decentralized applications.
ADA	Cardano	Proof-of-stake blockchain emphasizing scalability and formal verification.
SOL	Solana	High-throughput blockchain designed for low-latency transaction processing.
XRP	Ripple (XRP Ledger)	Payment-oriented digital asset for cross-border settlement.

Daily RV Observations and Overlap

Although intraday data are sampled at 5-minute intervals, the forecasting target is daily realized volatility. Table 2 therefore reports the number of *daily RV observations*, together with the effective sample period and validation checks for each asset.

Table 2. Daily realized volatility observations and sanity checks by cryptocurrency.

Coin	Daily RV Obs.	Period Covered	Missing Days	Missing Values
ADA	2787	2018-05-16 – 2025-12-31	0	0
BTC	3030	2017-09-15 – 2025-12-31	0	0
ETH	3030	2017-09-15 – 2025-12-31	0	0
SOL	1940	2020-09-09 – 2025-12-31	0	0
XRP	2770	2018-06-02 – 2025-12-31	0	0

No missing calendar days or internal missing values were detected within the effective daily RV samples after preprocessing.
Since the full historical availability differs across assets, a common overlap interval is identified for multi-asset analysis:

- **Common start date:** 2020-09-09
- **Common end date:** 2025-12-31

This common window ensures that pooled evaluation metrics and cross-sectional comparisons are computed over identical time spans.

4. Experiments

Although the original HAR(3) specification is constructed using rolling averages over the past 1, 5, and 22 trading days—corresponding approximately to daily, weekly, and monthly horizons in equity markets—the 24/7 trading nature of cryptocurrency markets suggests an alternative horizon structure. In continuous markets without weekend closures, a more natural decomposition is given by 1, 7, and 30 days, representing daily, weekly, and monthly dynamics under a calendar-time perspective.

Accordingly, the HAR model used in this study is specified as:

$$\mathrm{RV}_{t+1} = \beta_0 + \beta_d \mathrm{RV}_t^{(d)} + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+1}, \tag{3}$$

where

$$\mathrm{RV}_t^{(d)} = \mathrm{RV}_t, \tag{4}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{7}\sum_{i=0}^{6} \mathrm{RV}_{t-i}, \tag{5}$$

$$\mathrm{RV}_t^{(m)} = \frac{1}{30}\sum_{i=0}^{29} \mathrm{RV}_{t-i}. \tag{6}$$

This adaptation preserves the heterogeneous autoregressive structure while aligning the temporal aggregation with the continuous trading environment of cryptocurrencies.

TBA: A brief global summary of all experimental procedures and comparisons will be included after the complete empirical evaluation.

[1] A cite will be useful here.

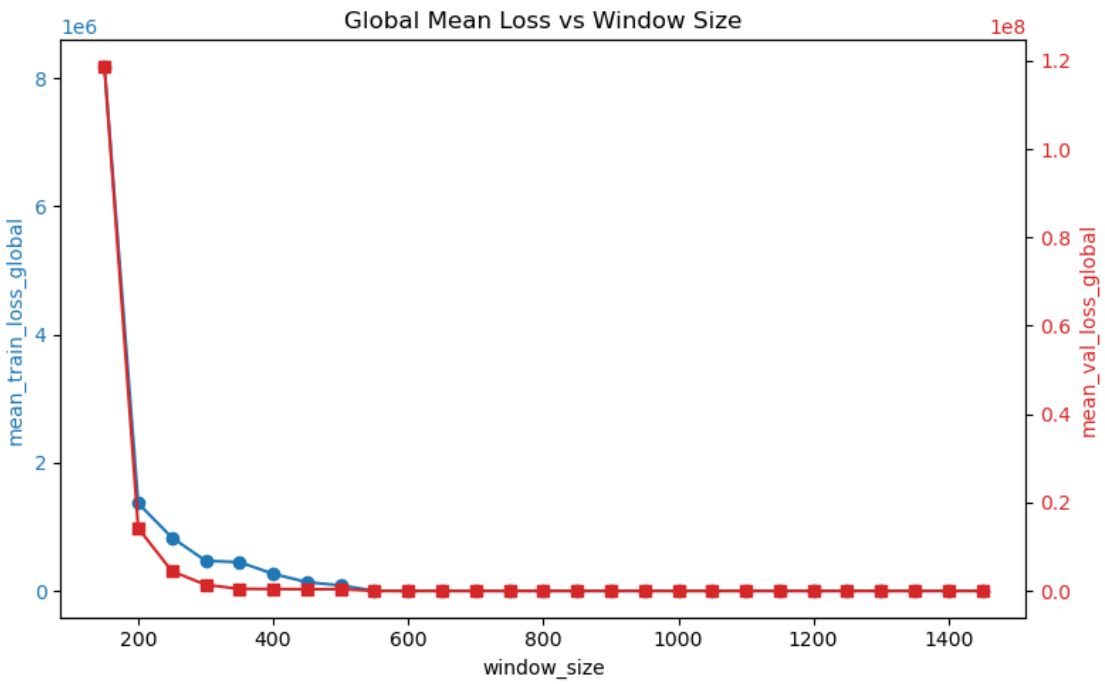

Figure 1. Mean training and validation QLIKE losses across fitting window sizes (integrated overlap sample).

4.1. Testing Sensitivity of the Fitting Window

The HAR(3) model parameters are estimated via Ordinary Least Squares (OLS). A key design choice concerns the size of the fitting (training) window. Since realized volatility is available at a daily frequency, increasing the number of observations requires extending the historical estimation window. However, although volatility exhibits long-memory properties, it is also well documented to be regime-dependent. Consequently, overly long estimation windows may dilute the relevance of recent observations by incorporating outdated regimes, while excessively short windows may lead to unstable parameter estimates.

This trade-off is particularly relevant in cryptocurrency markets, which are characterized by higher volatility and speculative dynamics relative to traditional financial markets. Standard assumptions derived from equity markets may therefore not transfer directly.

To investigate this issue, we implement a rolling window procedure with:

- A fixed validation window of 100 daily observations,
- A training (fitting) window of size
$$N = 50m, \quad m = 1, \dots, 30,$$
i.e., window sizes ranging from 50 to 1500 observations.

For each window size, the HAR parameters are estimated using OLS. Model performance is evaluated using the QLIKE loss, reported separately for training and validation sets. Within each rolling configuration, losses are averaged across all windows.

The initial analysis is conducted over the common overlap period shared by all cryptocurrencies, in order to ensure comparability and robustness. Figure 1 displays the mean training and validation losses as a function of the window size.

The results indicate that losses decrease as the window size increases, but only up to a certain point. Beyond that threshold, further enlarging the estimation window does not improve performance and may even slightly deteriorate validation accuracy.

Table 3 reports the window size achieving the minimum average loss in the integrated analysis.

Table 3. Optimal fitting window (integrated overlap sample).

Coin	L	Mean Train Loss	Mean Val Loss	Mean Loss
Integrated	1250	0.372367	0.352530	0.362449

A similar analysis is conducted individually for each cryptocurrency using its maximal available sample. Selecting the window size L that minimizes the average of training and validation losses yields the summary in Table 4.

The results consistently contradict the naive hypothesis that "more data is always better." Instead, an intermediate window length appears to balance the benefit of larger samples against the loss of local regime relevance. Across assets, the optimal window size typically lies between approximately 700 and 1200 daily observations, with 1250 observations in the integrated analysis.

An additional empirical observation—although not reported in tabular form—is that window sizes below 200 observations produce substantially larger losses. This finding underscores the importance of sufficient sample size for stable parameter estimation and generalization.

From a practical perspective, these results suggest that the HAR model for cryptocurrency volatility forecasting should be estimated using approximately two to three years of historical daily data. Such a horizon appears to provide a suitable compromise between statistical stability and responsiveness to evolving market regimes. This range balances the need for sufficient observations to ensure reliable parameter estimation with the necessity of preserving sensitivity to regime shifts and structural changes characteristic of cryptocurrency markets.

Table 4. Optimal fitting window by cryptocurrency.

Coin	L	Mean Train Loss	Mean Val Loss	Mean Loss
Integrated	1250	0.372367	0.352530	0.362449
BTC	1150	0.375448	0.383566	0.379507
ETH	900	0.320906	0.364735	0.342820
ADA	800	0.323731	0.382044	0.352887
SOL	1200	0.279160	0.260309	0.269735
XRP	700	0.428394	0.490552	0.459473

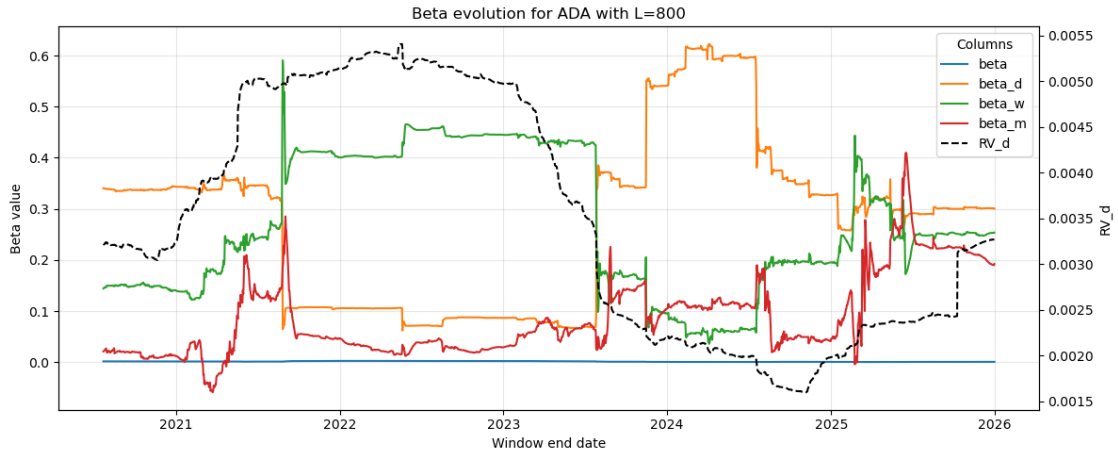

Figure 2. Evolution of HAR coefficients for ADA ($L = 800$).

4.2. Evolution of β Values Over Time

Once the optimal fitting window L is determined, the remaining object of analysis concerns the behavior of the HAR model parameters themselves. If the selected window length effectively balances statistical stability and responsiveness to local regimes, then the estimated coefficients should provide meaningful information about how the HAR structure interprets volatility dynamics over time.

To investigate this aspect, we proceed as follows. For each cryptocurrency, the previously identified optimal window size is used within a rolling estimation framework. At each step of the rolling procedure, the HAR parameters are estimated via OLS and stored. In parallel, a rolling window with the same horizon L is applied to the realized volatility series in order to compare the evolution of the coefficients with the contemporaneous level and structure of volatility.

Figures 2–6 report the temporal evolution of the intercept β, the daily coefficient β_d, the weekly coefficient β_w, and the monthly coefficient β_m for ADA, BTC, ETH, SOL, and XRP, respectively.

Several stylized patterns emerge.

First, the intercept β remains close to zero across all assets and time periods, suggesting that most of the predictive structure is captured by the heterogeneous lag components rather than by a constant drift term.

For ADA (Figure 2), realized volatility initially increases, then decreases, and eventually stabilizes around an intermediate level. During high-volatility phases, the weekly coefficient β_w tends to exceed the daily coefficient β_d, while during declining volatility phases the opposite relation is observed. The monthly coefficient β_m fluctuates over time without a clear monotonic trend, indicating variable long-horizon influence.

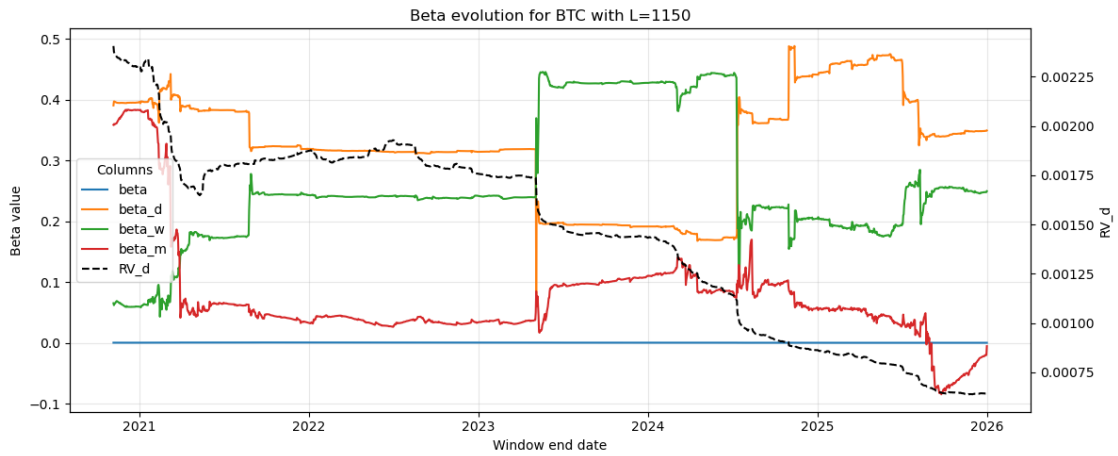

Figure 3. Evolution of HAR coefficients for BTC ($L = 1150$).

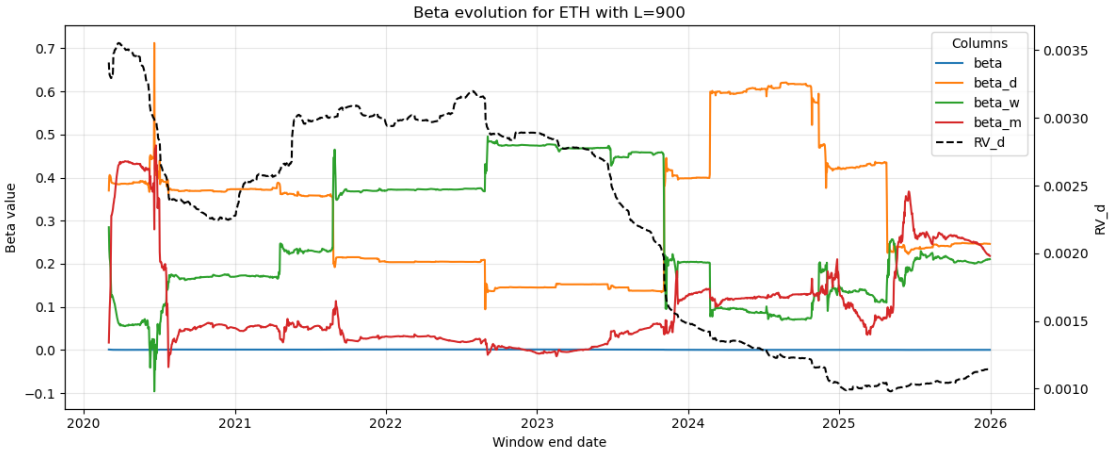

Figure 4. Evolution of HAR coefficients for ETH ($L = 900$).

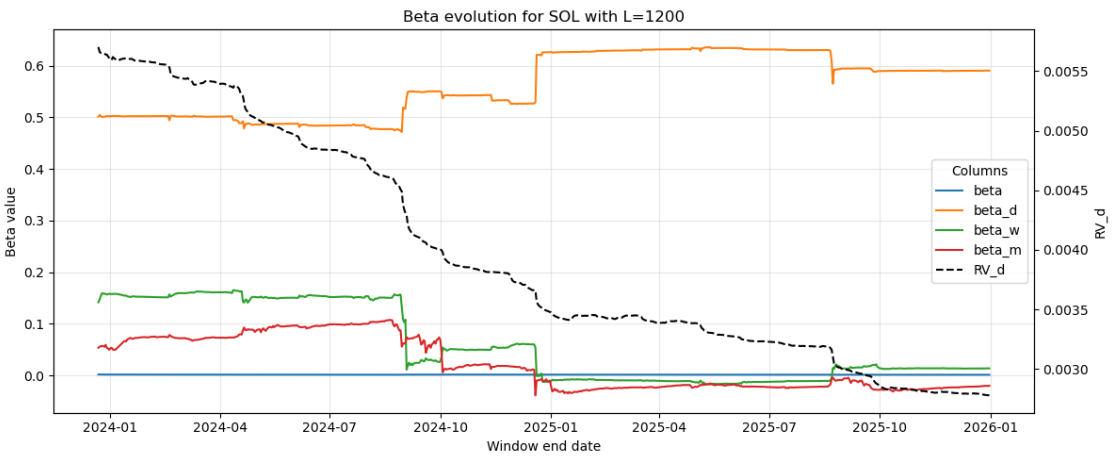

Figure 5. Evolution of HAR coefficients for SOL ($L = 1200$).

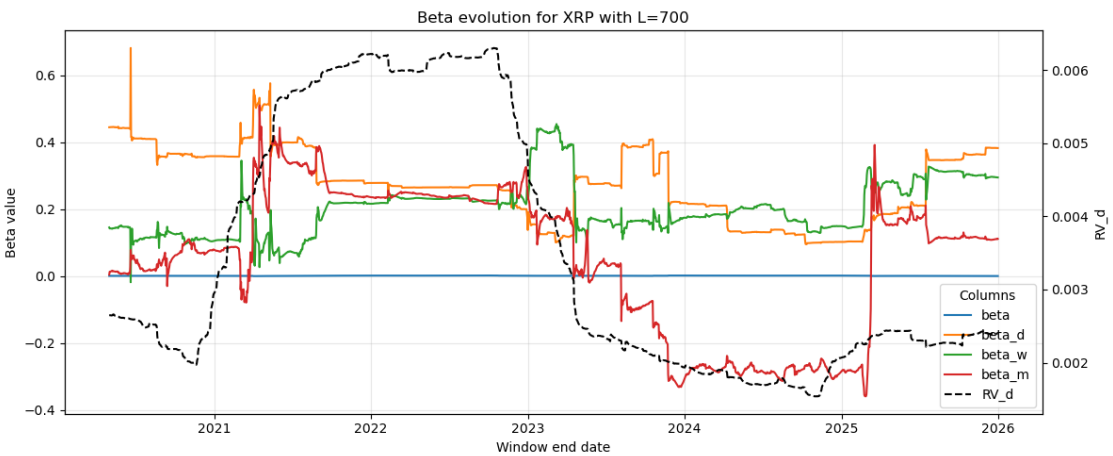

Figure 6. Evolution of HAR coefficients for XRP ($L = 700$).

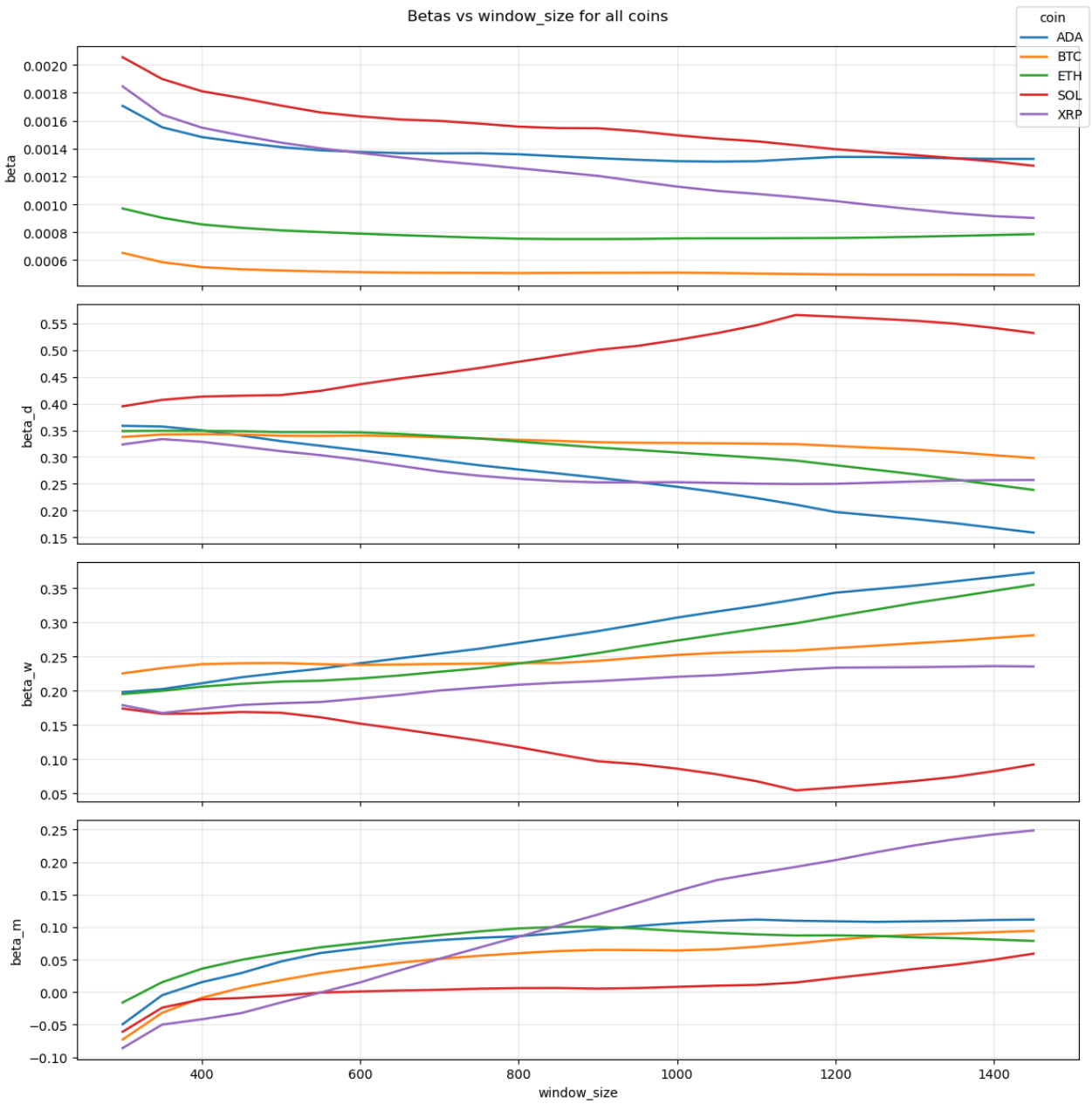

Figure 7. Average HAR coefficients as a function of fitting window size L.

BTC (Figure 3) exhibits a general downward trend in realized volatility over the sample. A persistent trade-off between β_d and β_w is observed: when one increases, the other decreases. Unlike ADA, the monthly coefficient β_m becomes negative toward the end of the sample, suggesting that the long-horizon component may occasionally act as a corrective force rather than a reinforcing one.

ETH (Figure 4) displays behavior broadly similar to ADA, with alternating dominance between daily and weekly components depending on the volatility regime.

SOL (Figure 5), estimated with the longest optimal window ($L = 1200$), shows comparatively stable coefficients. The monthly and weekly coefficients remain close to zero throughout most of the sample, while the daily coefficient β_d consistently dominates. This suggests that SOL volatility is primarily driven by short-horizon dynamics within the HAR framework.

XRP (Figure 6), with the shortest optimal window ($L = 700$), presents volatility dynamics similar in shape to ADA but with more stable β_d and β_w coefficients. In contrast, β_m exhibits erratic fluctuations, including both positive and negative values, indicating unstable long-horizon influence.

It is important to note that the apparent profile of realized volatility depends on the selected fitting window. Longer horizons tend to smooth volatility and emphasize broader trends, whereas shorter windows preserve sharper regime shifts. Consequently, the coefficient dynamics reflect not only intrinsic market behavior but also the observational scale at which the HAR model is estimated.

Mean Coefficient Behavior Across Window Sizes

As a complementary analysis, the experiment is repeated for window sizes ranging from $L = 300$ to $L = 1500$. For each window length, the mean value of each coefficient during the rolling estimation process is computed. Figure 7 summarizes the relationship between window size and average coefficient magnitude.

Several systematic patterns can be identified. As the fitting window expands, the intercept β tends to decrease in magnitude, reinforcing

the notion that heterogeneous lag components absorb most of the predictive content in larger samples.

Across most cryptocurrencies, the daily coefficient β_d decreases as the window length increases, while the weekly coefficient β_w exhibits the opposite tendency. This indicates a gradual redistribution of explanatory weight from very short horizons toward intermediate horizons as more historical data are incorporated.

The monthly coefficient β_m generally increases with window length, particularly in the case of XRP. This suggests that longer estimation horizons allow the model to capture more persistent, long-memory components through the monthly regressor. In this sense, the HAR structure effectively reallocates explanatory power across horizons as the fitting window expands: short-term influence diminishes, intermediate influence stabilizes, and long-horizon persistence becomes more prominent.

An exception to this pattern is SOL, where the daily component remains dominant even for large window sizes. This indicates that SOL volatility, within the HAR framework, appears more strongly driven by short-term dynamics than by longer memory components.

Overall, these findings highlight that the heterogeneous structure of the HAR model is not static but adapts to both the underlying asset and the chosen estimation horizon. The interaction between fitting window length and coefficient allocation provides additional insight into how volatility persistence is distributed across temporal scales in cryptocurrency markets.

5. Discussion

This section interprets the empirical results presented in the previous section. The goal is not to repeat the numerical findings, but to explain their economic and statistical implications and to connect them with the existing literature.

5.1. Sensitivity of HAR performance to the estimation window

Intermediate estimation windows appear to provide the best forecasting performance. This result can be interpreted through the classical bias–variance tradeoff in parameter estimation. Very short windows contain limited information and may lead to unstable parameter estimates, while very long windows incorporate observations from past market regimes that may no longer be representative of current market conditions.

In cryptocurrency markets, which are characterized by rapid structural changes, this tradeoff becomes particularly relevant. The empirical results suggest that windows between approximately 700 and 1200 observations provide a balance between estimation stability and adaptability to recent market dynamics.

Some key points in the literature are worth mentioning. [5] reports two types of estimations: one using the full sample (more than 3000 observations) and another based on rolling windows for out-of-sample forecasting. In the latter case, a rolling window of size 1000 is employed, which is similar in magnitude to the optimal windows found in this study.

Similarly, [8] explores practical aspects of HAR implementation. While their analysis focuses on transformations and estimation procedures, the size of the rolling window is not explicitly investigated. Instead, a window of 1000 observations is adopted, following previous implementations such as [6]. This suggests that the choice of window size has often been treated as a practical convention rather than as an empirical question.

A similar approach appears in [10], where a rolling window of 1220 observations is used without a detailed justification. The results of the present study provide empirical evidence that such window sizes may indeed correspond to near-optimal configurations.

Regarding the economic interpretation of the optimal window length, the results indicate that windows corresponding to approximately two to three years of data tend to perform well. One possible explanation is that cryptocurrency markets are often described as evolving through medium-term cycles. Although the precise mechanisms behind these cycles remain debated, the presence of such dynamics may explain why windows shorter than two years fail to capture persistent volatility patterns, while much longer windows incorporate outdated regimes.

5.2. Time-varying importance of HAR horizons

One of the most striking findings of the rolling regressions is the near-zero intercept observed across most estimation windows. This suggests that the level of volatility is largely explained by the lagged realized volatility components rather than by a constant term.

The HAR structure appears particularly effective in this context. The daily component captures the most recent information, the weekly component provides a smoother representation of short-term dynamics, and the monthly component reflects the broader volatility level. Together, these components provide a flexible representation of volatility dynamics without requiring a large intercept term.

Another important observation is that the relative importance of the daily and weekly coefficients varies over time. In some assets, such as ADA and ETH, the weekly component appears to become more relevant during periods of high volatility, possibly because it smooths the short-term fluctuations captured by the daily component. In calmer periods, the daily component may provide a more accurate estimate of volatility changes.

However, this pattern is not universal across all assets. For example, BTC and XRP do not exhibit the same clear alternation between daily and weekly dominance, and SOL shows a markedly different behavior. This suggests that the dynamics captured by HAR coefficients may depend on asset-specific characteristics.

A further observation concerns the relationship between the estimation window and the role of the monthly component. As the window size increases, the monthly coefficient tends to become more important, while the intercept and the daily coefficient decrease. This behavior is consistent with the well-known long-memory property of financial volatility. Larger estimation samples allow the model to capture persistent volatility structures more clearly.

In Figure 7, the monthly coefficient initially appears negative for very short windows but becomes positive as the window length increases. The negative values observed in short windows may reflect temporary mean-reversion effects or unstable parameter estimates. Once the window exceeds approximately 700 observations, the coefficient stabilizes at positive values, supporting the interpretation of the monthly component as a persistent baseline level of volatility.

5.3. Differences across cryptocurrencies

An additional finding of the analysis is that different cryptocurrencies exhibit distinct HAR coefficient structures. While several assets share similar dynamics, others display noticeably different behaviors.

For instance, SOL shows a strong dominance of the daily component, while the weekly and monthly components remain relatively small. This behavior contrasts with that observed in other cryptocurrencies, where longer horizons tend to play a more relevant role. Such differences may reflect variations in market maturity, liquidity, trading activity, or investor composition across assets.

5.4. Interaction between window size and horizon importance

The results also highlight an important interaction between the estimation window size and the relative importance of the HAR horizons. As the window size increases, the contribution of the monthly component becomes more prominent, while the daily component gradually loses importance.

This pattern suggests that the interpretation of HAR coefficients depends on the estimation horizon. When the estimation sample is short, the model is more sensitive to recent volatility fluctuations. In contrast, longer estimation windows allow the model to identify more persistent volatility structures.

Empirically, the coefficients appear to stabilize once the window size approaches approximately 1000 observations, providing a clearer interpretation of the roles of the daily, weekly, and monthly components.

5.5. Implications for volatility modeling

These results provide several implications for volatility forecasting in cryptocurrency markets. First, the HAR model remains a useful baseline framework for capturing heterogeneous volatility dynamics. However, the empirical evidence suggests that the model parameters should be interpreted dynamically rather than as fixed quantities.

Figures 2, 3, 4, and 6 illustrate how the coefficients evolve as the estimation window moves forward in time. As new observations enter the estimation sample and older observations leave it, the coefficients adjust to reflect the current volatility regime.

This dynamic behavior is particularly relevant in cryptocurrency markets, which are characterized by high liquidity and strong speculative activity. Understanding how HAR coefficients evolve over time may therefore provide useful insights for both forecasting performance and economic interpretation.

The findings also motivate extensions of the HAR model that allow for more flexible dynamics. Several approaches have been proposed in the literature, including regime-switching models [12], [14], [15], adaptive coefficient models [7], [13], [16], [23], and hybrid machine-learning frameworks [17], [20], [21], [22]. These approaches aim to capture nonlinearities and structural changes that may not be fully represented in the standard HAR specification.

6. Conclusion

This paper investigates the role of the estimation window size in the implementation of the Heterogeneous Autoregressive (HAR) model for realized volatility forecasting in cryptocurrency markets. While the HAR framework has been widely used in the literature, the choice of the rolling window size is often treated as a practical implementation detail rather than an empirical question.

The empirical analysis shows that the forecasting performance of the HAR model is sensitive to the length of the estimation window. Across the cryptocurrencies analyzed, the optimal rolling window is generally found to lie between approximately 700 and 1200 observations, corresponding to roughly two to three years of daily data. This result is consistent with commonly used window sizes in previous studies, although those choices were rarely justified through systematic evaluation.

The analysis of rolling coefficients also reveals important dynamic features of the HAR structure. In most cases, the daily and weekly components alternate in importance over time, while the monthly component captures the broader level of volatility persistence. As the estimation window increases, the monthly coefficient becomes more stable and increasingly relevant, reflecting the long-memory nature of financial volatility.

The results also highlight heterogeneity across cryptocurrencies. While several assets display similar volatility dynamics, others exhibit distinct coefficient structures, suggesting that the relative importance of volatility horizons may depend on asset-specific characteristics such as market maturity, liquidity, and trading activity.

Overall, the findings suggest that the estimation design plays a non-negligible role in the empirical performance and interpretation of HAR models. In particular, the choice of the rolling window size can influence both the forecasting accuracy and the relative importance of the daily, weekly, and monthly volatility components.

Future research may extend this analysis in several directions. Possible extensions include robustness checks using alternative HAR specifications such as HAR-J or HAR-Q models, the use of alternative estimation techniques such as weighted least squares, and the incorporation of adaptive or online-learning approaches that allow the model parameters to evolve more smoothly over time. Another promising avenue is the integration of high-frequency microstructure information or machine learning techniques to capture nonlinear dynamics in cryptocurrency volatility.

7. Acknowledgements

This research was conducted as part of a broader investigation into the dynamics of the HAR model and the role of historical information in the estimation of volatility models. The study was developed in parallel with the author's master's thesis research.

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